EXPERIMENT NO. 2: SIGNAL OPERATIONS AND TRANSFORMATIONS

OBJECTIVE 1:

To perform amplitude scaling and time scaling of continuous-time and discrete-time signals and analyze their effects on signal characteristics.

PROCEDURE:

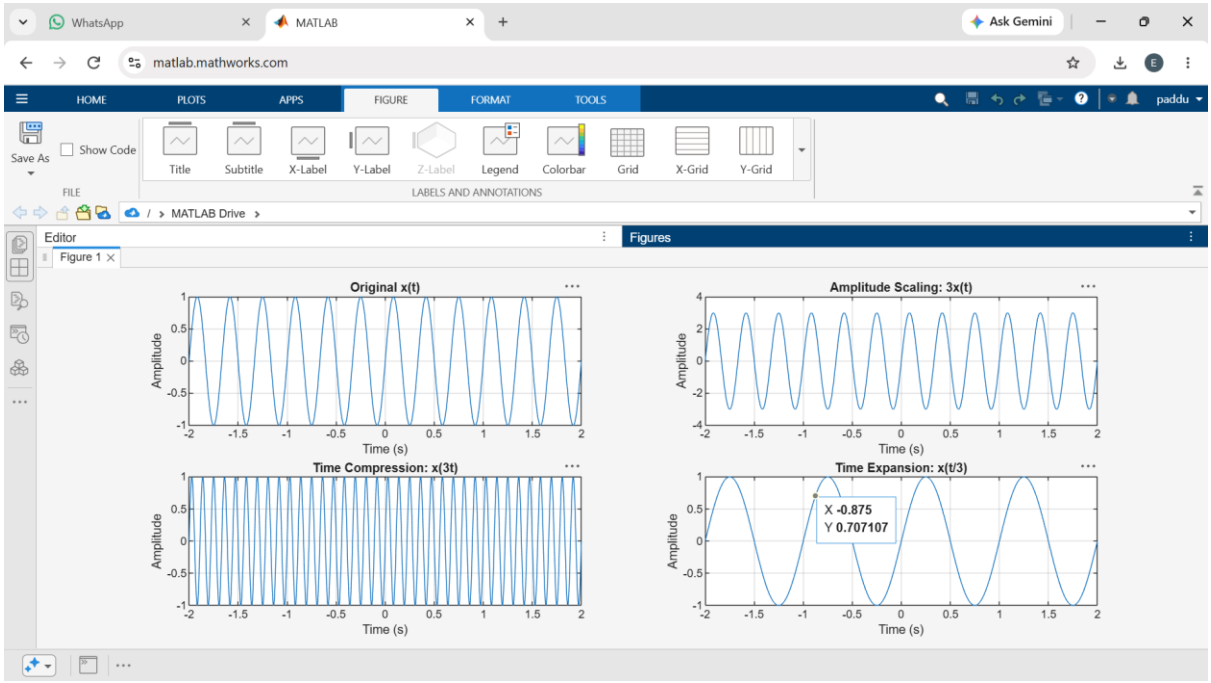
1. Generate the base signal $x(t) = \sin(2\pi 5t)$.
2. Apply amplitude scaling using $A = 2$.
3. Apply time compression using $x(2t)$.
4. Apply time expansion using $x(t/2)$.
5. Plot the original and transformed signals using common axes.
6. Measure the amplitude and period of the transformed signals.
7. Compare the measured results with the theoretical scaling relationships.

PROGRAM:

```
% Experiment 2(i): Amplitude and Time Scaling
clc;
clear;
close all;
t = -2:0.001:2;
x = sin(2*pi*3*t);
yA = 3*x;
yC = sin(2*pi*3*(3*t));
yE = sin(2*pi*3*(t/3));
figure;
subplot(2,2,1);
plot(t,x);
grid on;
title('Original x(t)');
xlabel('Time (s)');
ylabel('Amplitude');
subplot(2,2,2);
plot(t,yA);
grid on;
title('Amplitude Scaling: 3x(t)');
```

```
xlabel('Time (s)');
ylabel('Amplitude');
subplot(2,2,3);
plot(t,yC);
grid on;
title('Time Compression: x(3t)');
xlabel('Time (s)');
ylabel('Amplitude');
subplot(2,2,4);
plot(t,yE);
grid on;
title('Time Expansion: x(t/3)');
xlabel('Time (s)');
ylabel('Amplitude');
```

OUTPUT:



OBJECTIVE 3:

To perform addition and multiplication of two discrete-time sequences and analyze the resulting signal characteristics

PROCEDURE:

1. Define the sample index n.

2. Enter x[n] and y[n] with aligned indices.

3. Compute the sum sequence.

4. Compute the product sequence using pointwise multiplication.

5. Plot the input and resulting sequences using stem plots.

6. Compare the calculated samples with manual/theoretical results

PROGRAM:

```
% Experiment 2(iii): Addition and Multiplication

clc;

clear;

close all;

n = 0:4;

x = [2 3 1 4 2];

y = [1 2 3 1 2];

zSum = x + y;

zProd = x .* y;

figure;

subplot(2,2,1);

stem(n,x,'filled');

grid on;

title('x[n]');

xlabel('n');

ylabel('Amplitude');

subplot(2,2,2);

stem(n,y,'filled');

grid on;

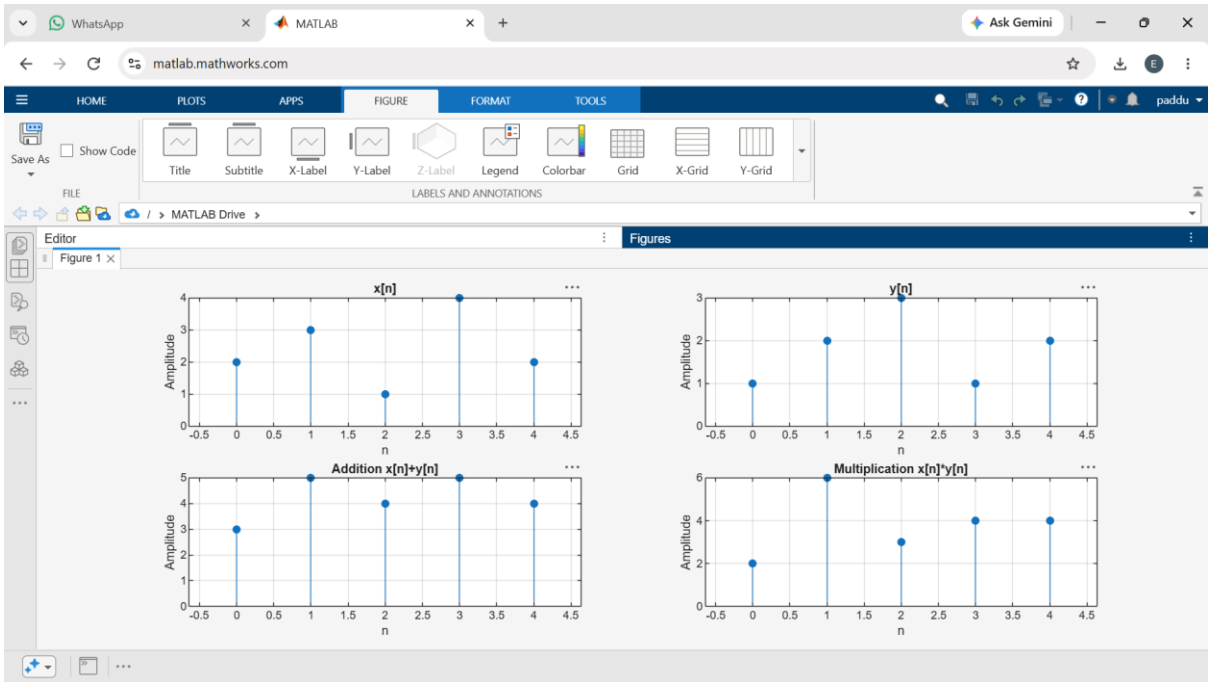
title('y[n]');

xlabel('n');

ylabel('Amplitude');
```

```
subplot(2,2,3);
stem(n,zSum,'filled');
grid on;
title('Addition x[n]+y[n]');
xlabel('n');
ylabel('Amplitude');
subplot(2,2,4);
stem(n,zProd,'filled');
grid on;
title('Multiplication x[n]*y[n]');
xlabel('n');
ylabel('Amplitude');
```

OUTPUT:



LABORATORY PERFORMANCE ASSESSMENT RUBRICS:

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage	Marks (10)	Marks obtained
(i) Perform amplitude scaling and time scaling of continuous-time and discrete-time signals and analyze their effects on signal characteristics.	**Amplitude & Time Scaling Implementation (MATLAB)**	Generates the original and scaled versions correctly for CT and DT with correct index handling. Uses plot for CT and stem for DT, with proper time and index vectors. Original and scaled plots are placed in subplots with matching axes, titles, axis labels and grid.	Generates most scaled signals correctly, with minor errors in scaling factors, index handling, axis limits or labelling.	Generates only basic scaled signals, with noticeable errors in code, time or index vectors, or plots, and needs assistance.	Unable to generate the scaled signals or produce meaningful plots.	**25**	2.5	
(i) Perform amplitude scaling and time scaling of continuous-time and discrete-time	**Analysis of Scaling Effects**	Correctly explains that amplitude scaling changes amplitude and energy but not duration	Explains most effects correctly, with minor errors or limited justification.	Gives basic observations on the effects. The explanation is incomplete and	Unable to analyze or explain the effects of scaling.	**25**	2.5	

signals and analyze their effects on signal characteristics.		or period. Explains that time compression shortens duration and period (T/a) and raises frequency, that expansion does the opposite, and that a < 0 reverses time. Explains that DT scaling discards samples and needs interpolation. Supports every point with waveform evidence.		needs guidance.				
(iii) Perform addition and multiplication of two discrete-time sequences and analyze	**Addition & Multiplication Implementation (MATLAB)**	Adds and multiplies two sequences correctly even when their index ranges differ.	Performs both operations correctly, with minor errors in index alignment, padding,	Performs the operations on simple equal-length sequences only, with noticeable	Unable to perform the operations or produce meaningful stem plots.	**25**	2.5	

the resulting signal characteristics.		Aligns the sequences on a common index vector with zero padding, and uses element-wise operator. Plots both inputs and the results with stem, correct n-axis, titles, axis labels and grid.	operators or labelling.	errors in code or plots, and needs assistance.				
(iii) Perform addition and multiplication of two discrete-time sequences and analyze the resulting signal characteristics.	**Analysis of Resulting Signal Characteristics**	Correctly determines the amplitude, the length or range of n, and the energy of the sum and product. Explains the periodicity of the result (for example, the sum is periodic with period	Analyzes most characteristics correctly, with minor errors or limited justification.	Gives basic observations on the resulting signals. The analysis is incomplete and needs guidance.	Unable to analyze or interpret the resulting signals.	**25**	2.5	

		equal to the LCM of the two periods). Explains how multiplication acts as gating or modulation. Interprets the results clearly with waveform evidence.						
					Total	**100**	**10**	